

Skymarket

The codename for a Windows mobile application store

Aviary Phoenix

An online image editor that offers everything from basic image retouching to complex effects. Definitely worth a look

From the Labs

And 10x10 is pretty coarse – if you imagine dividing your screen into a 10x10 grid, that's roughly a square inch. You're not going to have buttons [Ed. notes: on your screen] that size. So you're already at 100 servo motors for a not-very-high resolution. That's pretty much the state of the art in tactile displays right now. There's nothing that'll give you per-pixel resolution on a, say, 1024 x 768 resolution, where each pixel can be actuated.

There are things in the pipeline: there's a great project from Nokia [Ed. notes: *Haptikos*, which first surfaced in 2007], where you have micro-fluid chambers push up on a per-pixel level, but I haven't seen the technology public yet, and I believe it's fairly expensive at present.

d Where do you see your touchscreen technology being used?

CH: Unfortunately, the big limitation is that it doesn't

let you have dynamic buttons – you have to “pre-layout” the tactile buttons. So you can do maybe four or five interfaces, depending on how complicated you want to make the display. For our ATM display, we'd get about 25 different combinations. But interfaces that want to support four or five different screens – like an ATM screen, which has a number keypad

and some buttons in fixed places – that's where it's going to be a big win, because you can augment the display in a very simple way. It's not going to work fantastically well on your desktop machine or your laptop, because there are so many interfaces that it needs to support. If you wanted to have every button and widget on your desktop to be tactile, you'd have to have buttons everywhere, and that's just

not practical. If you wanted to augment just the start button in Windows, if it's in a fixed location, you could do that. But there's less of a win there.

A typical car [Ed. notes: in the US] might have GPS, radio, air conditioning, an entertainment system – there's only four or five different states that you could ever want to see. You could build a display that supports a unique tactile

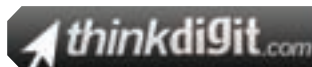
layout for each one of those. Those are definitely the most compelling applications.

Where they make a huge difference is driving. As you can imagine, your eyesight is consumed by trying to keep on the road. There's been a tendency in car manufacturers to move to touchscreens, because they want to support so many features. The big problem is that with touchscreen buttons with no

tactile feedback, you'd have to look at the screen every single time – you can't do it by feel, like you would with a physical dashboard. And that's a big problem – I wouldn't be surprised if accident rates increased because of this technology.

d Coming back to the ATM example – what are the restrictions of today's ATM interfaces? We've always thought they were pretty effective.

CH: They are pretty effective, although... we're young – people with limited eyesight have a much tougher time. Even when I stand directly in front of ATM interfaces with touchscreens, sometimes it still doesn't recognise my input for my PIN. For us, [Ed. notes: a tactile display] is not such a big deal, except that we might be able to type faster with less error. But for people who are slower, or who have less motor control, this could be a big win.

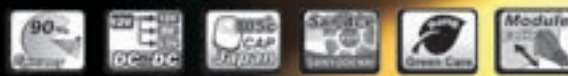


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